TOP2B-Related Immunodeficiency and Congenital Anomalies Spectrum (TICAS): A Comprehensive Disease Dossier

Disease: TOP2B-Related Immunodeficiency and Congenital Anomalies Spectrum (TICAS) **Causal gene: TOP2B** (DNA topoisomerase IIβ; HGNC:11997; chromosome 3p24.2; gene OMIM *126431) MONDO: MONDO:0012243 | OMIM: 609296 Synonyms: B-cell immunodeficiency–limb anomaly–urogenital malformation syndrome (BILU syndrome); Hoffman syndrome; Ablepharon-macrostomia-like syndrome (TOP2B form) Category:** Mendelian, autosomal dominant, ultra-rare (~14 reported patients)

Summary

The disease addressed in this report is caused by **TOP2B**, the gene encoding **DNA topoisomerase IIβ (topo IIβ)**, an essential nuclear enzyme that relieves topological stress during DNA replication and gene transcription. Despite the disease name containing synonyms (BILU, Hoffman, ablepharon-macrostomia-like syndrome) that could be mistaken for other conditions, all are unified under a single molecular etiology — heterozygous, predominantly dominant-negative variants in TOP2B. Çepni et al. (2023) proposed the unifying term **"TOP2B-related Immunodeficiency and Congenital Anomalies Spectrum (TICAS)"** to encompass these overlapping phenotypes (PMID: 37068767).

TICAS is an **autosomal-dominant, syndromic inborn error of immunity** defined by a clinical triad of: (1) B-cell-predominant immunodeficiency (agammaglobulinemia with absent/reduced circulating B cells, plus impaired NK-cell cytotoxicity), (2) distal/acral limb anomalies, and (3) urogenital malformations, with variable facial dysmorphism and neurodevelopmental involvement. The landmark discovery by Broderick et al. (2019) established TOP2B as the causal gene and demonstrated, using yeast, knock-in and knockout mouse models, that patient mutations act through a **dominant-negative effect** on enzyme function, blocking B-cell development (PMID: 31409799).

Mechanistically, TOP2B is required both for lymphocyte development and for the transcription of long developmental genes. The enzyme operates as a homodimer that transiently cleaves one DNA duplex to pass another through, and patient variants poison this strand-passage cycle in a dominant-negative fashion. Because complete loss of Top2b is perinatally lethal in mice, viable human disease arises from partial-function dominant-negative missense alleles rather than null alleles. A genotype-phenotype boundary exists: variants in the **TOPRIM domain and adjacent regions** produce TICAS, whereas the recurrent N-terminal **p.His63Tyr (ATPase domain)** variant produces a clinically distinct, severe neurodevelopmental disorder (PMID: 42963833). Management is entirely supportive — immunoglobulin replacement, infection prophylaxis, surgical correction of anomalies, and developmental/neurologic care — with no curative or disease-specific therapy.

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1. Disease Information

Overview. TICAS is an ultra-rare Mendelian disorder in which germline heterozygous variants in **TOP2B** cause a syndromic, inborn error of immunity combined with congenital structural anomalies. The disease was first delineated molecularly in 2019 when sequencing of patients with B-cell-restricted immunodeficiency revealed autosomal-dominant TOP2B mutations (PMID: 31409799). Broderick et al. note: *"TOP2B encodes a type II topoisomerase, an essential gene required to alleviate topological stress during DNA replication and gene transcription, with no previously known role in B cell development."*

Key identifiers:

Resource	Identifier
MONDO	MONDO:0012243
OMIM (disease)	609296
OMIM (gene TOP2B)	*126431
HGNC (gene)	HGNC:11997
Gene symbol	TOP2B

Synonyms / alternative names: B-cell immunodeficiency–limb anomaly–urogenital malformation syndrome (BILU syndrome); Hoffman syndrome; Ablepharon-macrostomia-like syndrome (TOP2B form); TOP2B-related Immunodeficiency and Congenital Anomalies Spectrum (TICAS, the preferred unifying term).

Information source. Information is derived from **aggregated disease-level resources and individual case reports/case series** in the primary literature (~14 patients total across all reports), supplemented by functional studies in yeast, mouse, zebrafish, and patient-derived iPSC models. There is no large EHR cohort; this is an ultra-rare condition.

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2. Etiology

Primary cause — genetic. TICAS is caused by **heterozygous germline pathogenic variants in TOP2B**, predominantly missense variants acting through a **dominant-negative mechanism** (PMID: 31409799; PMID: 37068767). There is no environmental, infectious, or toxic cause; recurrent infections are a *consequence* of the immunodeficiency rather than an etiologic trigger.

Genetic risk factors. The causal variants are the sole known risk factor. Variants cluster in the **TOPRIM domain and adjacent regions** for TICAS. No susceptibility loci or polygenic contributors have been described (the disorder is monogenic and ultra-rare).

Environmental / protective factors. None identified. Given the Mendelian, high-penetrance nature of the disorder, environmental modifiers and protective alleles have not been characterized. No gene-environment interactions have been reported.

Modifier considerations. Variable expressivity across reported patients (mild-to-severe) suggests the existence of modifiers, but none have been mapped. The specific TOP2B residue affected is itself the strongest determinant of phenotype (see Section 4 and Section 9).

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3. Phenotypes

TICAS presents a **triad** of immunodeficiency, acral/limb anomalies, and urogenital malformations, with variable facial dysmorphism and neurodevelopmental features (PMID: 37068767): *"three overlapping phenotypes characterized by immunodeficiency, acral and urogenital anomalies: Hoffman, BILU and Ablepharon-macrostomia-like syndrome."*

Phenotype	Type	Characteristics	Suggested HPO term
Agammaglobulinemia / reduced immunoglobulins	Laboratory abnormality	Congenital; severe; stable; core feature	HP:0004432 (Decreased circulating total IgG) / HP:0004313 (Decreased B cell count)
Absent/reduced circulating B cells	Laboratory abnormality	Congenital; B-cell developmental block	HP:0010976 (B lymphocytopenia)
Reduced NK-cell number/ cytotoxicity	Laboratory abnormality	Congenital; impaired cytotoxic function	HP:0040218 (Reduced natural killer cell count)
Recurrent infections	Clinical sign	Childhood onset; variable severity	HP:0002719 (Recurrent infections)
Distal/acral limb anomalies	Physical manifestation	Congenital; variable	HP:0002817 (Abnormality of the upper limb) / HP:0001167 (Abnormality of finger)
Urogenital malformations	Physical manifestation	Congenital; variable	HP:0000119 (Abnormality of the genitourinary system)
Facial dysmorphism	Physical manifestation	Congenital; variable	HP:0001999 (Abnormal facial shape)
Developmental delay / neurodevelopmental involvement	Clinical sign / behavioral	Variable; more severe with specific alleles	HP:0001263 (Global developmental delay)

Onset. All structural and immunologic features are **congenital/neonatal**; infections become apparent in infancy/childhood as maternal antibody wanes. **Severity** ranges from mild to severe across patients. **Progression** of structural anomalies is stable (non-progressive); immunodeficiency is lifelong.

Quality of life impact. Lifelong immunoglobulin replacement, infection susceptibility, and surgical needs for limb/urogenital anomalies substantially affect daily functioning. Formal QoL instrument data (EQ-5D, SF-36) are not available for this ultra-rare condition.

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4. Genetic / Molecular Information

Causal gene. TOP2B (DNA topoisomerase IIβ), OMIM *126431, HGNC:11997, located on chromosome 3p24.2. TOP2B encodes a ~180-kDa type IIA DNA topoisomerase.

Pathogenic variants. - **Classification:** Heterozygous **pathogenic / likely pathogenic** missense variants (ACMG/AMP), typically *de novo* but with demonstrated vertical transmission (an affected mother and two affected sons; PMID: 37068767). - **Variant type:** Predominantly **missense**; functionally loss-of-function/dominant-negative at the protein level. - **Location:** TICAS-associated variants affect the **TOPRIM domain and adjacent regions**; the distinct severe neurodevelopmental phenotype is caused by the N-terminal ATPase-domain variant p.His63Tyr (PMID: 42963833). - **Functional consequence: Dominant-negative** effect on the homodimeric enzyme — *"patient mutations in TOP2B have a dominant negative effect on enzyme function, resulting in defective proliferation, survival of B-2 cells, causing a block in B cell development, and impair humoral function in response to immunization"* (PMID: 31409799). - **Allele frequency:** Variants are private/ultra-rare; absent or vanishingly rare in gnomAD, consistent with high penetrance and strong selection. - **Origin: Germline** (de novo or inherited). No somatic association with TICAS.

Key recurrent variant. TOP2B NM_001330700.2:c.187C>T, **p.(His63Tyr)** (equivalently p.His58Tyr in transcript NM_001068.2) is a recurrent *de novo* ATPase-domain variant associated with severe neurodevelopmental disorder rather than classic TICAS (PMID: 42963833; PMID: 31953910; PMID: 28343847). Hiraide et al. report: *"We found a de novo variant in TOP2B (NM_001330700.1:c.187C>T, p.(His63Tyr)), which is identical to the previous case. The clinical features of the two individuals with the c.187C>T variant overlapped."* (PMID: 31953910).

Modifier genes / epigenetics. No modifier genes mapped. TOP2B physically interacts with **MeCP2** to regulate transcription (PMID: 41009596); this positions TOP2B within chromatin-regulatory machinery, but no disease-specific epigenetic signature has been defined. No chromosomal abnormalities (aneuploidy, translocations) are associated.

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5. Environmental Information

No environmental, lifestyle, or infectious etiologic factors are implicated. TICAS is a purely genetic Mendelian disorder. Infections are downstream consequences of the primary immunodeficiency. No toxins, radiation, occupational exposures, diet, or microbial agents have been linked to disease causation.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **heterozygous dominant-negative missense variant in TOP2B** (TOPRIM/adjacent region for TICAS; ATPase-domain p.His63Tyr for the severe NDD branch) **arises** germline (de novo or inherited).
2. The mutant topo IIβ subunit **incorporates into the obligate homodimer** and **poisons** the ATP-driven strand-passage catalytic cycle (G-segment cleavage → T-segment passage → religation), *reducing* net topoisomerase throughput below the functional threshold (dominant-negative; PMID: 31409799; PMID: 16100112). *Inferred:* partial retained activity plus DNA-damage induction for p.His63Tyr — *"H63Y retains partial enzymatic activity and appears to function as a dominant negative variant that induces DNA damage"* (PMID: 42963833).
3. Reduced topo IIβ activity **impairs relief of topological stress** during DNA replication and transcription, which **branches** into two principal downstream effects:
4. **Branch A (immune):** Impaired proliferation/survival of developing B-2 cells **leads to** a **block in B-cell development** → absent circulating B cells → **agammaglobulinemia / impaired humoral response** (PMID: 31409799); and reduced peripheral mature NK cells with **impaired cytotoxicity** — *"Mature NK cells were reduced in the periphery of TOP2B*

knockin mice consistent with patient reports, with reduced cytotoxicity toward target cell lines" (PMID: 35063500).

5. **Branch B (developmental/neural):** Impaired transcription elongation of **long genes (>200 kb)** critical for development **results in** reduced expression of neurodevelopmental and tissue-patterning genes → **limb/urogenital malformations, facial dysmorphism, and neurodevelopmental involvement** (PMID: 23995680; PMID: 24463367).
6. In neurons specifically, reduced TOP2B **impairs activity-dependent DNA double-strand-break formation** at early-response-gene promoters (Fos, Npas4), **blunting** rapid stimulus-induced transcription (PMID: 36206766; PMID: 42911807), and **dysregulates Rho-GTPase (RhoA/ROCK2, Cdc42) and Nurr1 signaling**, impairing neurite outgrowth/axon guidance (PMID: 29695291; PMID: 21610027).
7. The convergent result is the **clinical triad** plus variable neurodevelopmental manifestations.

Molecular detail

Enzyme mechanism and protein dysfunction. Type IIA topoisomerases act as homodimers that *"catalyze transient cleavage of a DNA duplex (the G-segment) to yield a double-stranded gap through which another duplex (the T-segment) is passed"* (PMID: 16100112), orchestrated by ATP-driven opening/closing of molecular gates (cryo-EM of human Top2α, PMID: 34016969). Because the enzyme is an obligate dimer, a single mutant subunit can poison the complex — the structural basis of the dominant-negative effect. TOP2B is recruited with **nuclear hormone receptors (androgen/estrogen receptor)** to regulatory sites: *"The DSB can be mediated by the class II topoisomerase TOP2B, which is recruited with the androgen receptor and estrogen receptor to regulatory sites on target genes and is apparently required for efficient transcriptional activation of these genes"* (PMID: 21385925).

Long-gene transcription. *"Expression of long genes is also reduced after knockdown of Top1 or Top2b in neurons, highlighting that both enzymes are required for full expression of long genes"* and *"many high-confidence ASD candidate genes are exceptionally long and were reduced in expression after TOP1 inhibition"* (PMID: 23995680). This links topoisomerase-dependent transcription to neurodevelopmental disease genes.

Immune mechanism. The B-cell developmental block and NK-cell defects define the immunodeficiency (PMID: 31409799; PMID: 35063500).

Neuroimmune. TOP2B suppresses microglial pro-inflammatory cytokine release — *"topo IIβ overexpression decreased the release of key proinflammatory cytokines, suggesting an anti-inflammatory role. Conversely, the silencing of topo IIβ resulted in increased cytokine release"* (PMID: 42776333).

Activity-dependent DSBs. *"Neuronal activity induces topoisomerase IIβ (Top2B) to generate DNA double-strand breaks (DSBs) within the promoters of neuronal early response genes (ERGs) and facilitate their transcription"* and *"stimulus-dependent calcium influx through NMDA receptors activates the phosphatase calcineurin to dephosphorylate Top2B at residues S1509 and S1511, which stimulates its DNA cleavage activity and induces it to form DSBs"* (PMID: 36206766). Repair of these breaks by TDP2 is required for normal circuits: *"Deleting Tdp2 in excitatory forebrain neurons in vivo caused DSB accrual, ERG overexpression, and increased excitatory synaptic transmission and cortical circuit dysfunction"* (PMID: 42879281).

Suggested ontology terms. GO:0006265 (DNA topological change); GO:0006266 (DNA ligation); GO:0006357 (regulation of transcription by RNA Pol II); GO:0030183 (B cell differentiation); GO:0000724 (DSB repair via HR); GO:0007409 (axonogenesis). Cell types (CL): CL:0000236 (B cell), CL:0000623 (natural killer cell), CL:0000129 (microglial cell), CL:0000540 (neuron). CHEBI: CHEBI:15422 (ATP, enzyme cofactor).

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7. Anatomical Structures Affected

Organ / system level. - **Immune system (primary):** bone marrow B-lymphopoiesis, peripheral lymphoid tissue, NK-cell compartment (UBERON:0002405 immune system; UBERON:0002371 bone marrow). - **Limbs / skeletal (acral):** distal upper/lower limbs (UBERON:0002101 limb; UBERON:0002389 manual digit). - **Urogenital system:** kidneys, genital structures (UBERON:0000990 reproductive system; UBERON:0000081 urogenital region). - **Craniofacial:** facial dysmorphism (UBERON:0001456 face). - **Nervous system (variable):** cerebral/cerebellar cortex, retina in models (UBERON:0000955 brain; UBERON:0000966 retina).

Tissue / cell level. Hematopoietic lineage (B cells, CL:0000236; NK cells, CL:0000623), neurons (CL:0000540; retinal ganglion cells CL:0000740 in zebrafish model), microglia (CL:0000129). TOP2B is expressed in both proliferative and post-mitotic neuronal compartments of the human fetal telencephalon — *"In the cortex, TOP2B is expressed in cell nuclei in both proliferative (SVZ) and post-mitotic compartments (intermediate zone and cortical plate)"* (PMID: 26612825).

Subcellular level. Nucleus / nucleoplasm / chromatin (GO:0005634 nucleus; GO:0005694 chromosome). Topo IIβ is a chromatin-associated nuclear enzyme.

Lateralization. Limb and urogenital anomalies may be bilateral or asymmetric; no consistent lateralization reported.

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8. Temporal Development

Onset. Congenital. Structural anomalies (limb, urogenital, facial) are present at birth; immunodeficiency is present from birth but clinically manifests in infancy as recurrent infections once maternal IgG declines. Onset pattern is **chronic/insidious** for the immune features.

Progression. Structural anomalies are **stable/non-progressive**. The immunodeficiency is **lifelong and chronic**, requiring continuous management. Neurodevelopmental features (when present) follow a developmental course; in the severe p.His63Tyr branch, epileptic encephalopathy and cerebral/cerebellar atrophy with hypomyelination indicate a **progressive** neurologic process (PMID: 42963833).

Critical periods. Embryonic/fetal development is the critical window for the structural anomalies (irreversible once formed). The perinatal period and infancy are critical for immune management.

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9. Inheritance and Population

Inheritance. Autosomal dominant. Both *de novo* and vertically transmitted cases are documented — an affected mother transmitted the variant to two affected sons (PMID: 37068767).

Penetrance / expressivity. Apparently high penetrance with **variable expressivity** — the reported spectrum ranges from mild to severe. Çepni et al.: *"the patients reported herein along with the previously defined 11 represent a phenotypic spectrum from mild-to-severe immunological, acral and urogenital involvement, for which we propose the acronym 'TOP2B-related Immunodeficiency and Congenital Anomalies Spectrum (TICAS)'"* (PMID: 37068767).

Genotype-phenotype correlation. This is a defining feature:

Variant region	Example variant	Phenotype
TOPRIM domain & adjacent regions	(TICAS-associated LoF/ dominant-negative)	TICAS: immunodeficiency + acral + urogenital anomalies
N-terminal ATPase domain	c.187C>T p.His63Tyr	Severe neurodevelopmental disorder, epileptic encephalopathy, movement disorder

Ha et al. state: *"The phenotype associated with the H63Y variant differs greatly from the TOP2B-related Immunodeficiency and Congenital Anomalies Spectrum (TICAS) syndrome that has been reported with heterozygous loss-of-function variants affecting the TOPRIM domain and adjacent regions of TOP2B"* (PMID: 42963833).

Epidemiology. Ultra-rare — approximately **14 reported patients** total across all publications (PMID: 37068767). Precise prevalence/incidence cannot be calculated. No founder effect, no consanguinity requirement (dominant), no reported sex or ethnic predilection. Germline mosaicism is plausible but not documented.

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10. Diagnostics

Genetic testing (definitive). Diagnosis requires identification of a **heterozygous pathogenic TOP2B variant** by **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)**, or a targeted inborn-errors-of-immunity/primary-immunodeficiency gene panel including TOP2B. Single-gene TOP2B sequencing is appropriate when the phenotype is recognized. Chromosomal microarray/karyotype are generally unrevealing (no large structural lesions).

Immunologic laboratory workup. - Serum immunoglobulins (IgG, IgA, IgM) — **hypo-/ agammaglobulinemia**. - Lymphocyte immunophenotyping (flow cytometry) — **absent or markedly reduced circulating B cells**; reduced NK cells. - Functional NK-cell cytotoxicity assays — **reduced** (PMID: 35063500). - Vaccine-response (specific antibody) testing — impaired humoral response.

Imaging / structural evaluation. Skeletal imaging for limb anomalies; renal/urogenital ultrasound for urogenital malformations; brain MRI where neurodevelopmental features are present (may show cerebral/cerebellar atrophy and hypomyelination in the severe branch).

Differential diagnosis. Other agammaglobulinemias (e.g., BTK-related X-linked agammaglobulinemia, autosomal-recessive agammaglobulinemias) — distinguished by the **associated congenital limb and urogenital anomalies** and TOP2B genotype. Ablepharon-macrostomia syndrome (TWIST2) and other multiple-congenital-anomaly syndromes are differentials for the structural features.

Screening. No disease-specific newborn screening exists; TREC-based SCID screening would not reliably detect this B-cell/NK-cell-predominant defect. Cascade genetic testing of at-risk relatives is appropriate once a familial variant is known.

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11. Outcome / Prognosis

Survival/mortality. No formal survival statistics exist for this ultra-rare disorder. Prognosis is driven by infection control (immunodeficiency) and the severity of structural/neurologic involvement. With immunoglobulin replacement and infection prophylaxis, the immunodeficiency is manageable.

Morbidity. Significant lifelong morbidity from recurrent infections, limb/urogenital anomalies requiring surgery, and (in a subset) neurodevelopmental disability. The severe p.His63Tyr neurodevelopmental branch carries higher morbidity — medically refractory epileptic encephalopathy, hyperkinetic movement disorder, hypotonia, and progressive brain atrophy (PMID: 42963833).

Complications. Recurrent/severe infections, surgical complications, and developmental/neurologic sequelae.

Prognostic factors. The **specific variant/domain** is the strongest prognostic determinant (TOPRIM-region TICAS vs. ATPase-domain severe NDD), along with the degree of immunodeficiency and presence/severity of neurologic involvement.

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12. Treatment

No curative or disease-specific therapy exists. Management is **supportive and multidisciplinary** (NCIT:C70742 Supportive Care Intervention).

- **Immunoglobulin replacement therapy** (IVIG/SCIG) for agammaglobulinemia (NCIT:C564 Immunoglobulin Therapy).
- **Antimicrobial prophylaxis** and prompt treatment of infections (NCIT:C15844 Antibiotic Therapy).
- **Surgical correction** of limb and urogenital anomalies (timing individualized).
- **Developmental, neurologic, and rehabilitative care** — physical/occupational/speech therapy; antiepileptic management for seizures (particularly the severe neurodevelopmental branch).
- **Genetic counseling** for affected families (autosomal dominant, 50% transmission risk).

Advanced/experimental therapeutics. No gene therapy, cell therapy, RNA-based, or targeted therapy exists or is in trials for TICAS. Hematopoietic stem cell transplantation has not been established as standard. Pharmacogenomic considerations are not defined.

Caution. Because TOP2B is a target of topoisomerase-poison chemotherapeutics (e.g., etoposide, doxorubicin), such agents are mechanistically relevant to the enzyme but are **not** treatments for TICAS; conversely, the enzyme's dysfunction is central to the disease.

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13. Prevention

Primary prevention is not possible (genetic, congenital). Available measures are: - **Genetic counseling** and **prenatal/preimplantation genetic testing** for families with a known TOP2B variant (ACMG/ACOG frameworks). - **Cascade genetic screening** of at-risk relatives. - **Secondary/tertiary prevention:** early immunoglobulin replacement, infection prophylaxis, vaccination strategy adaptation (avoid live vaccines in significant immunodeficiency), and surveillance for complications.

No immunization, behavioral, environmental, or public-health prevention applies to disease causation.

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14. Other Species / Natural Disease

TICAS is a human Mendelian disorder; no naturally occurring equivalent disease is described in companion animals or wildlife (no OMIA entry for a spontaneous TOP2B syndrome). **Orthologous genes:** mouse *Top2b* (NCBI Gene), zebrafish *top2b*, and *S. cerevisiae TOP2* are functionally conserved and used experimentally. Evolutionary conservation of the strand-passage mechanism is high, enabling cross-species modeling. No zoonotic or transmission relevance (non-infectious genetic disease).

Taxonomy / NCBI Taxon: *Homo sapiens* (9606), *Mus musculus* (10090), *Danio rerio* (7955), *Saccharomyces cerevisiae* (4932).

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15. Model Organisms

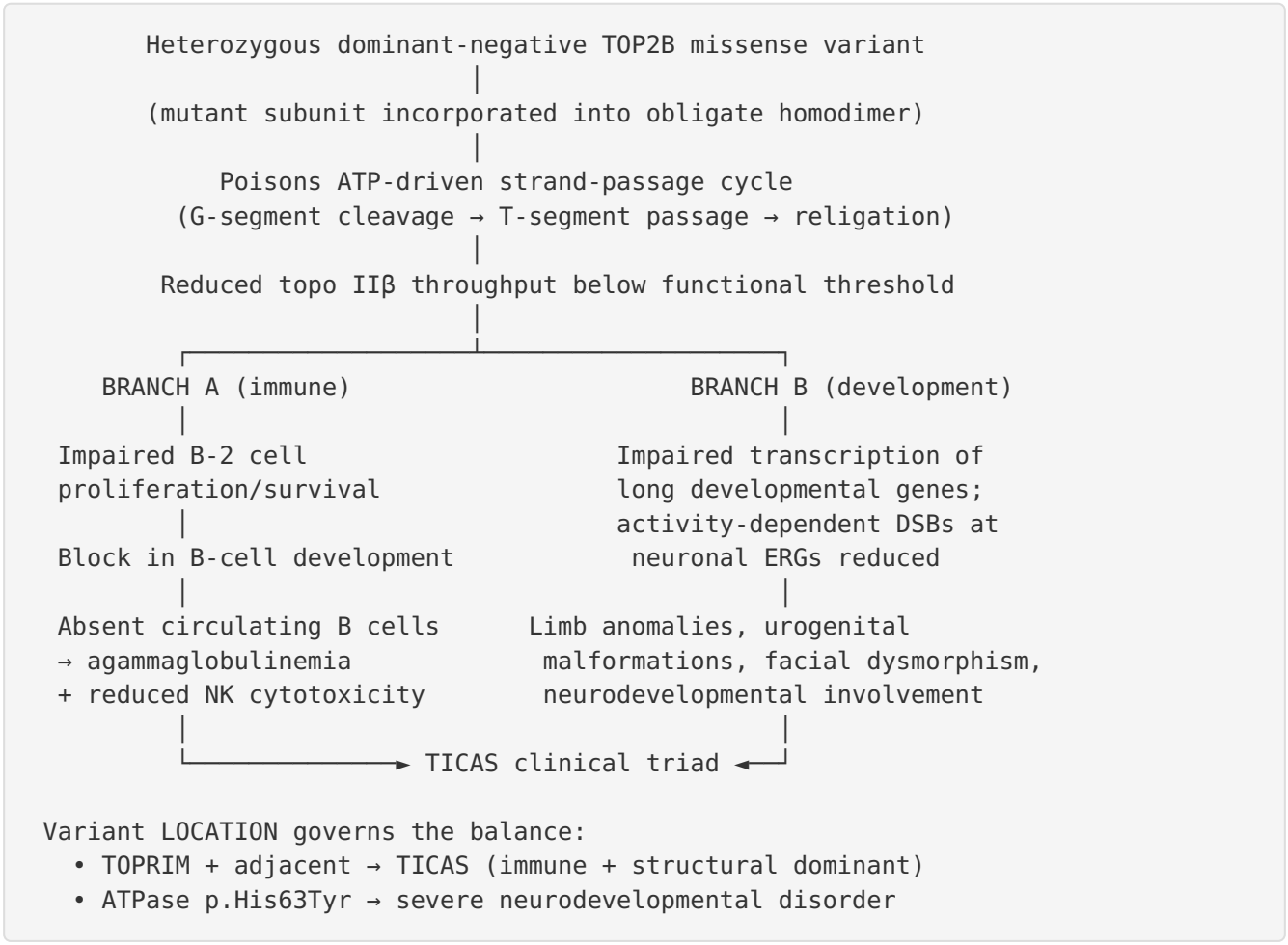
Cross-species models recapitulate distinct disease features, and the complete null is perinatally lethal — explaining why human disease arises from dominant-negative missense alleles.

Model	System	Key finding	PMID
S. cerevisiae (yeast)	Complementation/ overexpression	Patient alleles behave as dominant-negative; chimeric human H63Y lethal on overexpression	31409799, 42963833
Mouse knock-in (patient variant)	In vivo	Reproduces B-cell developmental block / reduced B-2 cells	31409799
Mouse knock-in	In vivo	Reduced peripheral NK cells with impaired cytotoxicity	35063500
Mouse constitutive / brain-specific KO	In vivo	**Perinatally lethal**	24463367
Mouse retina-specific KO	In vivo	Delayed neuronal differentiation, plexiform/ photoreceptor degeneration, cell loss	24463367
Zebrafish *top2b* "notorious" mutant	In vivo	Lamina-specific retinal ganglion cell axon/dendrite targeting defects	21610027
Patient-derived iPSC → NK cells	In vitro	Recapitulates reduced NK cytotoxicity	35063500
Human MSC neural differentiation; SH-SY5Y	In vitro	topo IIβ loss → Rho-GTPase dysregulation, impaired neurite outgrowth	29695291, 33402057

Broderick et al.: *"We use Saccharomyces cerevisiae, and knockin and knockout murine models, to demonstrate that patient mutations in TOP2B have a dominant negative effect on enzyme function"* (PMID: 31409799). Li et al.: *"Top2b deficiency resulted in delayed neuronal differentiation, degeneration of the plexiform layers and outer segment of photoreceptors, as well as dramatic reduction in cell number in the retina"* (PMID: 24463367).

Resources: MGI (mouse *Top2b*), ZFIN (zebrafish *top2b*), SGD (yeast *TOP2*).

Mechanistic Model / Interpretation



The unifying interpretation is that TOP2B is a **dosage-sensitive, dual-function enzyme**: it is indispensable for lymphocyte development (especially B cells) and for the elongation of transcription through long developmental genes and activity-dependent neuronal genes. A dominant-negative lesion lowers effective enzyme activity enough to disrupt these processes but not enough to be lethal (unlike a complete null, which is perinatally fatal in mice). The precise residue affected tunes which functions are most compromised, producing the TICAS-versus-neurodevelopmental genotype-phenotype split.

Evidence Base

PMID	Title (abbrev.)	Role
31409799	*Mutations in topoisomerase IIβ result in a B cell immunodeficiency*	Landmark: establishes causal gene, AD inheritance, dominant-negative B-cell block (yeast + mouse)
35063500	*Disease-associated mutations in topo IIβ result in defective NK cells*	NK-cell defect in knock-in mice + patient iPSC-NK cells; names Hoffman syndrome
37068767	*Revisiting TOP2B-related phenotypes: three new cases*	Defines TICAS, unifies 3 syndromes, ~14 patients, vertical transmission
42963833	*Recurrent NDD with dominant-negative TOP2B H63Y*	Genotype-phenotype split; H63Y retains partial activity, induces DNA damage
23995680	*Topoisomerases facilitate transcription of long genes linked to autism*	Long-gene transcription mechanism
24463367	*Top2b required for retinal development/survival*	Neuronal requirement; perinatal lethality of null
16100112	*Nucleotide-dependent domain movement in Top2 ATPase*	Strand-passage catalytic mechanism
21385925	*Transcription-induced DSBs*	TOP2B + hormone receptor DSB-driven transcription
26612825	*TOP2A/TOP2B in fetal telencephalon*	Human neuronal expression
29695291	*Topo IIβ stimulates neurite outgrowth via Rho-GTPases/Nurr1*	Neurite/axon mechanism
21610027	*Topo IIβ required for RGC axon/dendrite targeting*	Zebrafish neural model
42776333	*Topo IIβ suppresses microglial cytokine release*	Neuroimmune regulation
36206766	*Calcineurin dephosphorylates topo IIβ; neuronal DSBs*	Activity-dependent DSB/ERG mechanism
42911807	*Programmed DNA breaks drive ERG potentiation*	ERG transcription mechanism

PMID	Title (abbrev.)	Role
42879281	*TDP2 repairs TOP2B-generated DSBs*	Downstream repair/circuit consequences
31953910	*De novo TOP2B in GDD/ASD*	Recurrence of His63 residue in NDD
28343847	*De novo TOP2B in GDD/ID*	His58 (alt transcript) in NDD
34016969	*Cryo-EM allosteric regulation of human Top2α*	Structural basis of allostery/gating
40656194	*Infantile epileptic spasms in TOP2B deficiency*	Expands neurologic phenotype
41009596	*MeCP2–TOP2β transcriptional consequences*	Chromatin-regulatory interaction

Limitations and Knowledge Gaps

1. **Ultra-rarity (~14 patients).** All clinical conclusions rest on small case series/reports; prevalence, penetrance, survival, and QoL cannot be quantified robustly.
2. **Mechanism inferences for structural anomalies.** The link between impaired long-gene transcription and specific limb/urogenital malformations is **inferred** from model systems, not directly demonstrated in patient tissues.
3. **Genotype-phenotype boundary not fully resolved.** Why TOPRIM-region variants spare the brain while p.His63Tyr devastates it — despite both being dominant-negative — is incompletely understood (differential residual activity, tissue-specific dosage sensitivity, or DNA-damage induction are candidate explanations).
4. **No disease-specific therapy or trials.** Treatment evidence is extrapolated from general immunodeficiency management.
5. **Epigenetic and multi-omic data are sparse.** The TOP2B–MeCP2 interaction hints at chromatin-level dysregulation, but no patient-derived transcriptomic/epigenomic signature exists.
6. **Modifier genes unmapped**, leaving variable expressivity unexplained.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** to aggregate the scattered ~14+ cases, enabling natural-history, penetrance, and genotype-phenotype analyses.
2. **Patient-derived iPSC differentiation into B-lineage, NK, limb mesenchyme, and neural organoids** to directly test the long-gene-transcription hypothesis for structural anomalies via RNA-seq (focus on genes >200 kb).
3. **Domain-specific knock-in mouse lines** (TOPRIM-region vs. p.His63Tyr) to dissect the genotype-phenotype split with matched immune, skeletal, urogenital, and neurologic phenotyping.
4. **Structure-function mapping** of TICAS variants on the human TOP2B homodimer (building on cryo-EM of Top2α) to correlate residual activity and DNA-damage induction with clinical severity.
5. **Assess topoisomerase-modulating or chromatin-directed therapeutic strategies** in cellular/animal models, and evaluate whether residual-activity variants are amenable to activity-restoring interventions.
6. **Characterize the TOP2B–MeCP2 axis** in patient neurons to determine whether TICAS shares transcriptional-dysregulation mechanisms with Rett/MECP2-duplication spectrum disorders.

Report compiled from 24 primary and model-organism studies across five investigation iterations. Evidence source types span human clinical case series, mouse/zebrafish/yeast models, patient-derived iPSCs, in vitro cell systems, and structural/computational analyses.
